DATA MINING I

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Group 66

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Wonderful Wines of the World

Project Report

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1. Exploratory data analysis (EDA)

1.1 General overview of WWW data

We are analysing a dataset containing 10.000 rows, each identifying a customer, from *Wonderful Wines of the World* (WWW). This dataset consists of 21 columns, including 19 numerical and 2 categorical features, that represent various types of information, ranging from variables related to the customer personal information to their purchasing preferences. After the initial examination, we found that there are no missing values or duplicates. Based on our preliminary analysis, we got these initial conclusions:

- The median age of the costumers is 48 years old, ranging from 18 to 78 years old.
- Income ranges from $10,000 to $140,628, with an average of $69,904.
- Customers spent on average 16.7 years at school.
- 41.9% have children under 13 years old living at home and 47% have one or more teens (13 to 19 years old).
- In the last 18 months, customers made an average of 14.6 wine purchases, with an average of 62.4 days since their last purchase, and they also spent an average total of 622 dollars.
- Customers appear less focused on discounts, with only 32.4% of purchases made at discounted prices. However, some customers exclusively purchased discounted products.
- 42% of purchases were done through the web, visiting the website an average of 5.2 times.

1.2 Wine Preferences

Analysing the wine most favoured by the costumers:

- **Dry red wines** are the most popular. For some customers, this is the only type of wine they buy. With the customer purchasing on average 50.4% of this type of wine.
- **Dry white wines** are the second most popular, making up to 28.5% of purchases on average. Along with dry red wines, these are the only two wine types purchased by all customers at least once.
- **Sweet white wines, Sweet red wines** and **dessert wines** have a smaller market, with 7.1%, 7.1% and 6.9% of purchases on average, respectively.

Our data suggests that Exotic wines are a special category within the wine categories, working as an indicator of the percentage of Exotic wines purchased by a customer. Exotic wines can be any of the previously mentioned wine types. On average, customers purchase 10% of Exotic wines, however, for some, this category constitutes the main part of their wine purchases.

1.3 Data Anomalies

During our research we also discovered some data anomalies that need to be carefully analysed:

- The income values start at 10.000$ as if a lower value were not allowed with 40 customers having exactly this has their income. This raises concerns about accuracy, for example, for customers (aged 18–22), they are likely to be still studying and may not have an income.

- We have customers with 18 years old that are customers for over a year, this would be forbidden by law.
- All customers have over 540 days of history of purchase, indicating that recent customers may have all been excluded.
- Some 18-year-old customers are recorded as having a "teen at home," which is biologically impossible.

Based on our discoveries, we can make a few assumptions:

- The income values and purchase history suggest that this dataset may represent a specific segment of customers, those with incomes of at least $10,000 and purchase histories with more than 540 days. This could be a case of *sampling bias*.
- We found 18-year-olds that are customers for more than one year, this should be marked, however, we won't perform any changes as they won't affect the outcome of our analysis.

1.4 Correlations

We can extract some additional insights by plotting the correlations between two variables using Pearson's correlation coefficient:

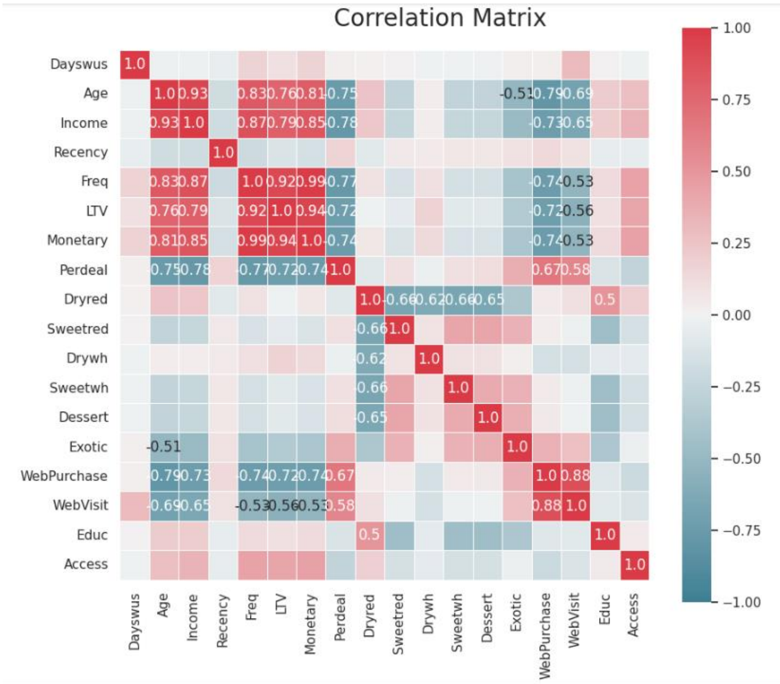

Fig. 1 – Correlation matrix, heatmap

- ***Frequency*** has a very high level of correlation with ***LTV*** and ***Monetary***, with 0.92 and 0.99. This would suggest multicollinearity. It will result in less reliable statistical inferences. Under the current circumstances and being impossible to collect more data under different conditions, the solution is to remove one of the columns. The same applies to ***Income*** ~ ***Age*** and ***LTV*** ~ ***Monetary***.
- We also observe negative correlations, ranging from -0.53 to -0.79, highlighting the correlations between **Age** ~ **WebPurchase** and **Perdeal** ~ **Income** with –0.79 and –0.78 respectively. Strong negative correlations can affect clustering, particularly for methods that rely on distance metrics

(e.g., **K-means clustering**, **Hierarchical clustering**, and **DBSCAN**). One way to address this issue is by applying **PCA**.

- There are some correlations between the wine categories, but none of them is extremely strong.

For other types of associations, we used the following metrics:

- **Cramer's V** for associations between categorical variables.
- **Point-Biserial Correlation** for associations between binary categorical variables and numerical variables.

Both tools are extensively explained in Annex 1.

The results were not statistically significant, with exception of Recency ~ Kidhome (0.76) and Webpurchase ~ Kidhome (0.6).

2. Data Preparation

2.1 Feature selection

One of the initial steps in data cleaning and preparation is selecting the best features in our dataset to maintain the relevance of each feature while reducing redundancies. This helps minimize bias, decrease complexity, and enhance the performance of clustering algorithms.

We have decided that the threshold for correlation should be above 0.8 (or below -0.8). In these cases, we consider the redundancy too high to retain the feature in the model. As a result, **Age**, **Monetary**, **WebPurchase**, and **CustID** were preliminarily eliminated:

- **Age** is highly correlated with **Income**, **Freq**, and **Monetary**, and it is also near the threshold of 0.8 with variables such as **Perdeal** and **WebPurchase**. Furthermore, **Age** raises concerns about reliability. Since WWW operates in the US market, where the legal drinking age is 21, we identified several customers below this age. Moreover, when combined with **Dayswus**, subtracting the age with period registered, we would find even more cases of customers registered below the majority age. That would make it impossible for the company to be complied with US laws.
- **Monetary** faces a similar redundancy issue as **Age**, being highly correlated with **Frequency**, **LTV**, and other variables. Logically, it is easy to understand its relationship: *"Customers who buy often tend to spend more."*
- **WebPurchase** is highly correlated with **WebVisit.** However, the issue extends beyond the correlation. From a business perspective, these variables are almost parallel: "*Customers who visit the website more frequently tend to make more online purchases". We* chose WebVisit instead, because WebPurchase has a higher collinearity with other variables.
- **CustID** was removed as it is simply an identifier and does not provide any descriptive information about customers.

The removal of those features does not imply that after the clustering they cannot be used for a better understanding of each cluster.

2.2 Data Transformation

2.2.1 Feature engineering/ Data Integration

Based on the data frame and the business problem descriptions, we had some insights to transform or create new features that can improve our understanding of customer types after clustering.

Access: The interval of accessories purchased by clients is between 0 and 3. Among the 1991 customers who bought accessories, 1579 purchased only one. Given that, we create a new binary feature (0/1) - indicating whether a customer purchased any accessory in the last 18 months.

KidHome and **TeenHome**: Although those features have an irrelevant correlation of -0.06, we understand that represent similar situations. Hence, we combined both into a new feature: **Minor_home,** which can provide useful information after clustering.

2.2.2 Anomaly detection – Outliers

Before normalizing the data, we manually removed some outliers. By looking into each histogram, we have identified that some of them had a right skewed distribution. So, to deal with this issue, we established a "floor"/"ceiling" for minimum/maximum values for each feature, inducing normality – where any value below/above would be equal the "floor"/"ceiling".

Capping of demographics and value features: **Frequency**: 46, **Recency**: 102, **LTV**: 1,200, **Perdeal**: 96.

Capping of wine features: **Sweetred**: 40, **Drywh**: max 65, min 2, **Sweetwh**: max 35, **Dessert**: max 35, **Exotic**: max 85.

2.2.3 Data Normalization

To enhance efficiency when comparing datapoints across multiple dimensions, we normalized all numerical variables, using the Min-Max formula. This scaled those data points between 0 a 1, preserving their statistical significance and enabling meaningful comparisons.

2.2.4 Outlier Detection Using DBSCAN

After normalizing and winsorizing the data, we will remove the outliers by using DBSCAN. First, we defined the EPS parameter using K-Distance graph. By looking to the graph, we see that the curves become more vertical at 0.6 – so that is the value for the EPS. For DBSCAN, we set:

- **min_samples** = 30
- **n_jobs** = 4
- **EPS** = 0.6

The algorithm and our chosen settings, removed 2.45%, preserving 9755 rows. With our initially defined threshold of 3% for max of outliers' removal, we found this result as satisfactory.

3. Segmentation Customer Value/Engagement

3.1 Features choice

For Customer Value/Engagement, we initially chose the following features:

1. Frequency: Frequency is highly correlated with Monetary, LTV, Income and Age. We chose to keep Frequency instead of the other 4 because it has the highest average correlation in between those variables.

2. Recency: Is an interesting variable when looking into value/engagement segments and has no high correlation to other variables.

3. Days with us: Is very interesting in order to understand the loyalty of the customer and also has no high correlation to other variables.

4. WebVisit: Is relevant because we want to understand the patterns of how our customers engage with us. We chose to use WebVisit instead of WebPurchase, as those two are highly correlated, but WebVisit is less correlated to other variables than WebPurchase.

5. PerDeal: With PerDeal we want to include this variable in the segmentation in order to differentiate which customers tend to buy products on offer and which don't. There are several medium-high correlations of this variable to other, but as none of them exceeds our defined threshold of 0.8 we decided to keep this variable for the segmentation.

6. Education: We want to understand if education plays a significant role when defining the value/engagement segments of our customers.

3.2 Determining the number of clusters

To define the number of k we used Elbow Curve Method and Silhouette Analysis, which resulted in different outputs.

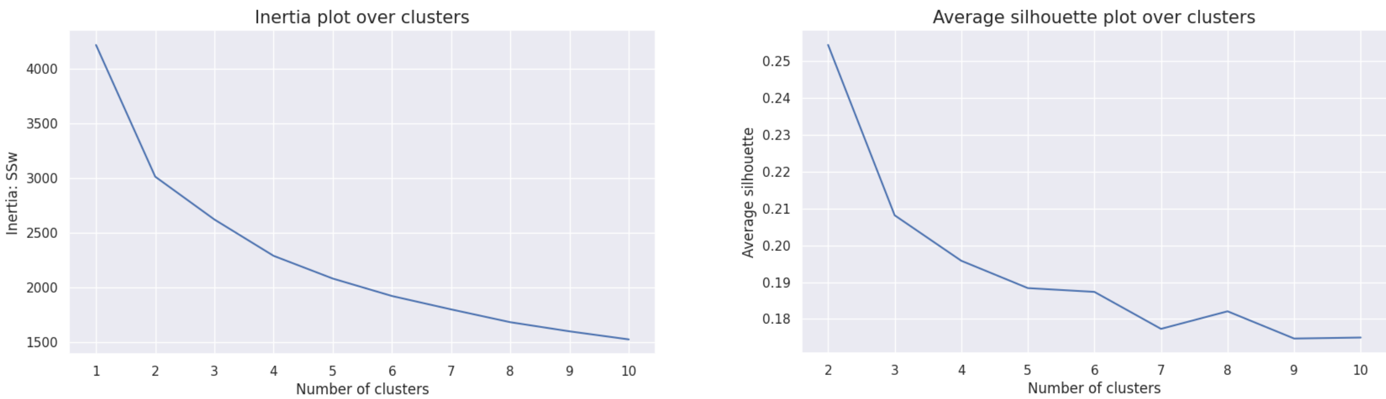

Fig. 2 – Elbow curve, Silhouette Score (Segmentation 1)

While the Elbow Curve has room for interpretation and shows “elbows” at k=2 & k=4, the Silhouette Analysis clearly states that 2 clusters are the best solution.

In order to understand better what is actually the best clustering solution for our business problem, we decided to try out 3 different k's:

- **K = 4**: While Freq, Recency, Dayswus, WebVisit & Perdeal have very different values in different clusters, the variable Educ shows very little variability. Based on the insights of this clustering solution, in order to further improve the segmentation by reducing dimensionality, we will remove this variable later and run the algorithm again. Furthermore, it is visible that cluster 1 & 2 are very similar to each other besides in the variable Recency.

km_labels_elb	Freq	Recency	Dayswus	WebVisit	Perdeal	Educ
0	0.456002	0.481546	0.248012	0.289078	0.086264	0.628045
1	0.092979	0.804620	0.480530	0.652718	0.572216	0.575010
2	0.115665	0.256985	0.508796	0.665330	0.554462	0.571336
3	0.588574	0.507134	0.769895	0.461807	0.098865	0.627237

Fig. 3 – K-means Segmentation 1, K=4

- **K = 2**: Educ shows a similar pattern as it did in the previous version with 4 clusters, having only very little variance between the clusters. Interestingly in this version with only 2 clusters also the mean of Recency and Dayswus show almost no difference between the clusters, while this variable showed high variability when having 4 clusters.

km_labels_silh	Freq	Recency	Dayswus	WebVisit	Perdeal	Educ
0	0.104682	0.535254	0.496116	0.658292	0.560703	0.572620
1	0.527964	0.486209	0.498128	0.367382	0.085651	0.629345

Fig. 4 – K-means Segmentation 1, K=2

- **K = 3:** This approach gives us somewhat unexpected results, as it looks like the two clusters with high Frequency from the solution with 4 clusters got merged into one (broadly speaking; of course, many data points were also categorized differently than in the other try for sure). Therefore, we are losing interesting differentiations about the variables Dayswus and Webvisit, which previously distinguished these 2 clusters significantly.

km_labels_3	Freq	Recency	Dayswus	WebVisit	Perdeal	Educ
0	0.116973	0.795880	0.495704	0.644950	0.533423	0.579486
1	0.566655	0.495677	0.494482	0.333939	0.065388	0.631471
2	0.141630	0.252930	0.501960	0.647049	0.501094	0.576570

Fig. 5 – K-means Segmentation 1, K=3

After all we can conclude that using 4 clusters for the final clustering is the best solution for us. Even if in this case 2 of the clusters only vary significantly in 1 variable, as the variable is Recency, this variable is a very crucial from business perspective and therefore justifies the separate cluster.

To get our final Customer Value/Engagement Segmentation we therefore ran the algorithm again with 4 clusters with the only difference of dropping the Educ variable which showed very little variance between the clusters. These are the results:

km_labels_elb_final	Freq	Recency	Dayswus	WebVisit	Perdeal
0	0.095765	0.804922	0.485632	0.653230	0.567517
1	0.589512	0.501116	0.768927	0.459720	0.097202
2	0.115013	0.257538	0.504034	0.664215	0.555082
3	0.458578	0.484078	0.246545	0.286730	0.084951

Fig. 6 – K-means final Segmentation 1, K=4

3.3 Visualization & Interpretation

In order to have a clearer understanding of each cluster, we performed new algorithms and created new graphs to visualize the composition of each cluster. For that we created a Line Graph for Cluster Means and a Bar Graph for each Cluster size.

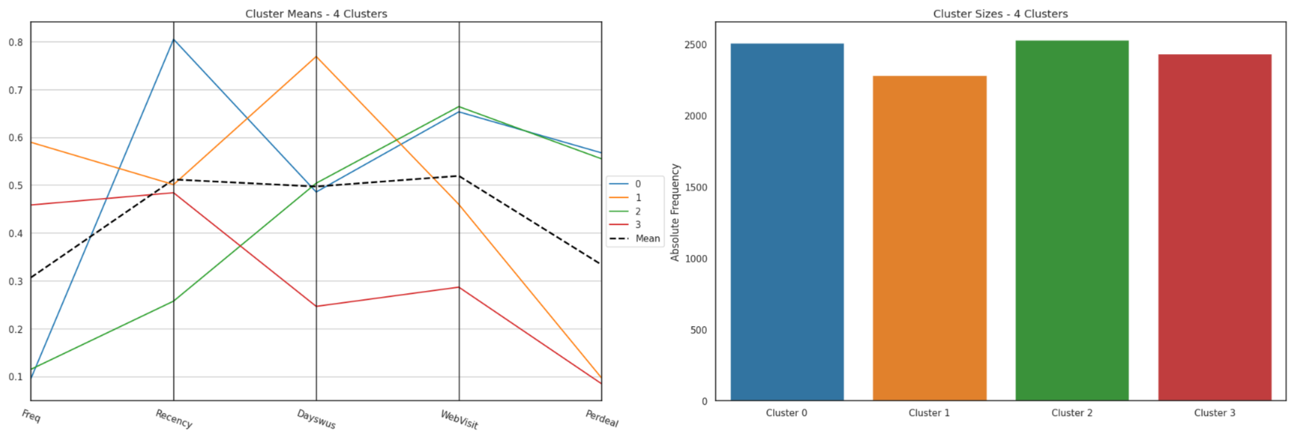

Fig. 7 – Segmentation 1 – line and bar charts

These visualizations allow for a great comparison between the clusters and for interpretation:

- **Cluster 0:** Represents customers that buy the least frequently, which also means they are on average the youngest, have the lowest income, the lowest Lifetime Value and spend the least money on our products (we can derive that due to the high correlation of these variables to Frequency). Those customers haven't made a purchase in a long time and the time they have been our customers is about average. They are a segment that visits our website a lot and therefore also does more web purchases than average. Also, those customers buy very often products that are on offer so we could conclude they are very price sensitive.

- **Cluster 1:** Represents customers that buy the most frequently, which also means they are on average the oldest, have the highest income, the highest Lifetime Value and spend the most money on our products. While the time that passed since their last purchase (Recency) is about average, those customers have been with WWW for quite some time and their visits our website are a bit less than average and therefore also their amount of web purchases is a bit less than average. Those customers buy very rarely products that are on offer, so we could conclude they are not very price sensitive.

- **Cluster 2:** Represents customers that buy less frequently than the average customer, which also means they are on average younger, have a lower income, a lower Lifetime Value and spend less money on our products. Those customers made their last purchase more recently than any other cluster and the time they have been with WWW is about average. They are the segment that visits our website the most and therefore also make the most web purchases. Also, those customers buy very often products that are on offer, so we could conclude they are very price sensitive.

- **Cluster 3:** Represents customers that buy more frequently than the average customer, which also means they are on average older, have a higher income, a higher Lifetime Value and spend more money on our products. While the time that passed since their last purchase (Recency) is about average, those customers are rather new clients at WWW and they are the segment that visits our website the least and therefore also does the least web purchases. Also, those customers buy very rarely products that are on offer, so we could conclude they are not very price sensitive.

Besides understanding the characteristics of each of the clusters, we can also see that the number of customers in each cluster is quite similar, which tells us that all of the 4 clusters represent a relevant amount of customers in our portfolio and are therefore worth to take into consideration.

3.4 Introducing cluster to the original Data Frame and Final Conclusions

Finally, we introduced the clusters to our original data frame. This allows us to examine the original data without Min-max normalization and to take into enhance our interpretation based on variables which were not part of the clustering.

km_labels_elb_final	Custid	Dayswus	Age	Educ	Income	Kidhome	Teenhome	Freq	Recency	Monetary	LTV	Perdeal	Dryred	Sweetred	Drywh	Sweetwh	Dessert	Exotic	WebPurchase	WebVisit	Access
0.000000	6055.33	889.94	35.20	16.64	49369.62	0.73	0.54	5.31	119.86	144.57	7.27	54.48	49.08	7.17	28.74	7.50	7.48	21.70	54.92	6.53	0.11
1.000000	6012.20	1088.25	61.55	16.98	92110.40	0.09	0.45	27.59	51.11	1320.17	490.28	9.33	54.69	5.85	28.11	5.64	5.70	9.75	30.15	4.60	0.40
2.000000	5923.40	902.82	35.83	16.62	50576.18	0.72	0.54	6.18	26.27	170.04	8.89	53.29	50.31	7.22	27.90	7.45	7.09	21.05	55.00	6.64	0.10
3.000000	6011.74	722.58	62.84	16.98	93232.45	0.07	0.38	21.64	49.39	970.49	375.79	8.16	51.29	6.49	29.67	6.45	6.07	9.02	26.80	2.87	0.36

Fig. 8 – Segmentation 1, average values

What do the results shown in this table tell us?

As we reintroduced the original DF again to our clusters, most of the interpretations made previously by only analyzing the 5 variables used for the final clustering hold true. This confirms the strong correlations between some of the variables in our dataset.

Some slight changes are visible when looking at the age or income variable for example. Linking the clusters back to the original dataset, where no treatment of one-dimensional outliers took place yet, we can see that not Cluster 1 has the highest average age and income, but Cluster 3. Even though those variables have a correlation of 0.83 and 0.87 to Frequency, this correlation was not enough to identify the cluster with the oldest customers and the highest income correctly based on Frequency. Knowing that the difference of those two clusters in the Frequency variable is not extremely high and being aware that there are other variables distinguishing those clusters much more (*Dayswus* & *WebVisit*), this insight is not a reason for us to doubt the quality of our clustering.

This analysis gave us good insights about the differences concerning customer value/engagement within our dataset. As a next step, we have to generate a better understanding of our customers' buying behaviour.

4. Segmentation Customer Behaviour

4.1 Features choice

For Customer Behaviour, we chose the 'Wine' Features. We understand that for Behaviour Segmentation, it is important to analyse how these features relate to one another. Although the 'non-wine' features can explain consumption in certain ways, we prefer to focus solely on the wine features to observe how different customers purchase various types of wine or combinations of 'baskets'. This also prevents us from having a too high dimensionality in our clustering.

4.2 Determining the number of clusters

As in Customer Value Segmentation, to define the number of k we used Elbow Curve Method and Silhouette Analysis, which resulted in different outputs:

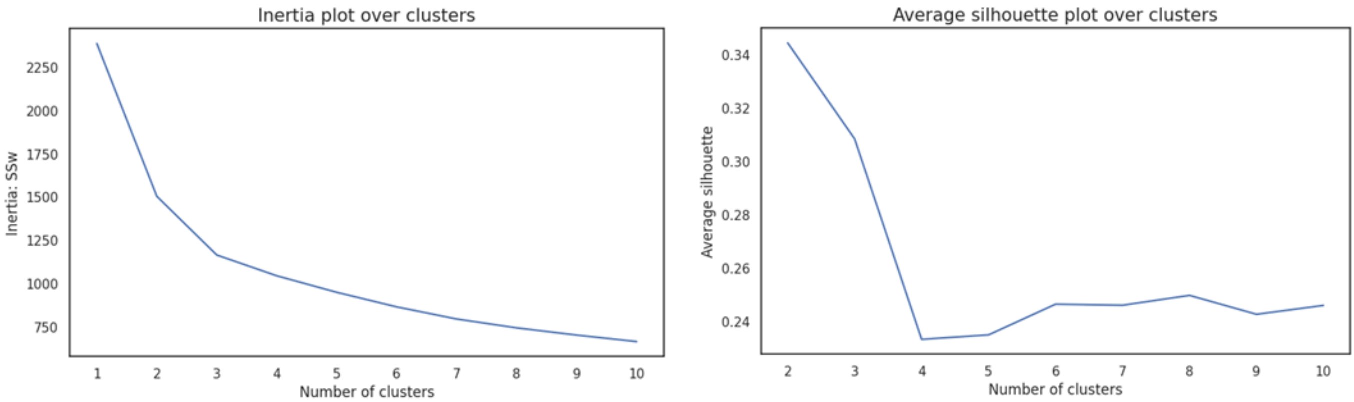

Fig. 9 – Elbow curve, Silhouette Score (Segmentation 2)

For the Elbow Curve Method, we could argue that 2, 3 and maybe 4 would be the optimum number of clusters. Although the Inertia curve has apparently the same slope from 4 and onwards. Looking into the silhouette, 2 seems to be the optimum number of clusters, although the fall between 2 and 3 is smaller than between 3 and 4.

To understand in practical terms what could be the best solution we tried 3 different K's analysis. Starting with k=4, then k=2 and finally k=3. After that we calculated the average of each feature in each cluster and compared them. Those were our results:

- **K = 4**: A segmentation of 4 clusters has shown to be too weak to proceed with. We could say that all clusters are too similar between themselves, in a way that we even could not consider them as cluster.

km_labels_elb2	Dryred	Sweetred	Drywh	Sweetwh	Dessert	Exotic
0	0.516713	0.160678	0.436840	0.181880	0.171092	0.105787
1	0.486163	0.181101	0.426986	0.213509	0.211979	0.257094
2	0.501208	0.179974	0.411936	0.211041	0.201362	0.248717
3	0.550673	0.144762	0.412809	0.158895	0.160890	0.114052

Fig. 10 – K-means Segmentation 2, K=4

- **K = 2**: This cluster solution has proven to be much more interesting. We now have two distinct clusters.
    - Cluster 1 has a stronger preference for Dry Red wines, while occasionally favouring Dry White wines. They clearly show less interest in sweeter wines or in the exotic ones.
    - Cluster 0, on the other hand, demonstrates a more balanced range of tastes. Although they tend to prefer Dry White wines, they purchase sweeter wines (including Dessert) more frequently than customers in Cluster 1 and are more likely to experiment with Exotic wines.

km_labels_silh2	Dryred	Sweetred	Drywh	Sweetwh	Dessert	Exotic
0	0.297128	0.278926	0.538949	0.326166	0.314221	0.243444
1	0.689770	0.075480	0.326618	0.082144	0.082575	0.133704

Fig. 11 – K-means Segmentation 2, K=2

- **K = 3:** By having 3 clusters we had even more interesting results.

We now have a clear new distinct Cluster 2. This cluster has a distinct taste for sweeter wines than 0 and 1. It's composed of customers that are more likely to try different type of wines (higher Exotic mean). On the other hand, this cluster is less likely to buy Dry Red Wines than other Clusters, but they are located between 0 and 1 when it comes to Dry White Wine Consumption.

Cluster 0 can be interpreted as less bold in terms of purchase - they prefer Dry Red Wine more than 1 and 2, and besides, **Dryred** they would go for Dry White Wine. They are sometimes open for sweet wines, but it is a minority of the cluster or a rare behaviour among them.

In the end, Cluster 1 – although closer to Cluster 0 – seems to be the most balanced group. They normally buy 3x more sweet wines than Cluster 0, but always less than half when compared with Cluster 2. They are more likely to buy **Dryred** than cluster 2 (2x) but much less than Cluster 0. When it comes to Dry White Wine consumption, they are the 'winners', being the top purchasers of this type. Finally, they are close to Cluster 0 when comparing Exotic wines.

km_labels_3wh	Dryred	Sweetred	Drywh	Sweetwh	Dessert	Exotic
0	0.745417	0.061864	0.263549	0.069943	0.067695	0.136150
1	0.414073	0.171004	0.577591	0.186354	0.190089	0.156731
2	0.200709	0.407862	0.421594	0.496214	0.462588	0.359058

Fig. 12 – K-means Segmentation 2, K=3

In conclusion, by clustering into 3 different segments, we aimed our goal of identifying a suitable number of clusters that are far distinct between themselves, exhibiting different behaviours in terms of wine purchases.

4.3 Other Visual Comparisons

In order to have a clearer understanding of each cluster, we performed new algorithms and created new graphs to visualize the composition of each cluster, as we already did for Customer Value Segmentation. For that we created a Line Graph for Cluster Means and a Bar Graph for each Cluster size.

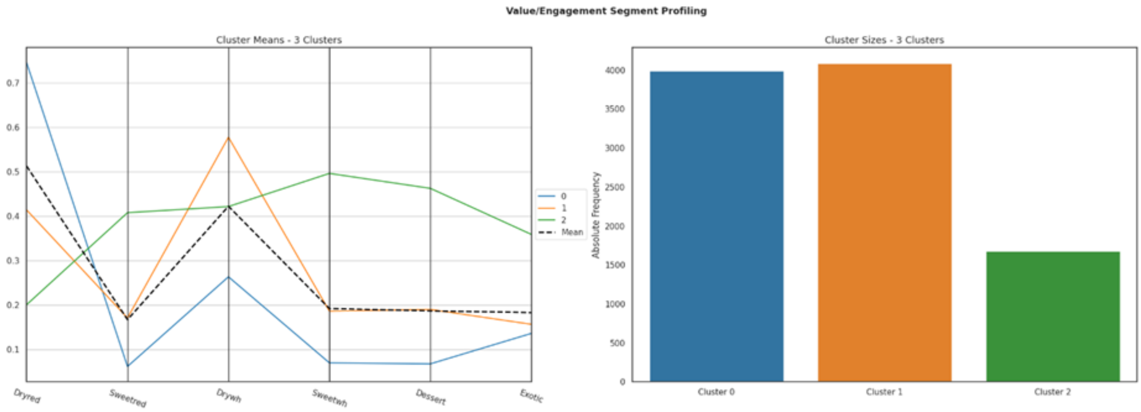

Fig. 13 – Segmentation 2 – line and bar charts

Some of our previous conclusions are now clear. Cluster 1 is the one who is closer to mean. It only fails when looking at Dry White Wine. As we can conclude that they are the most balanced among the 3 clusters.

Cluster 0, as we have seen before, shows a stronger preference for Dry Red Wine, than Dry White Wine and below average when looking for other types and exotics.

Cluster 2 is above the average when looking for preference for Sweet, Dessert and Exotic wines. And it is close to the average for Dry White Wines.

Cluster 1 and 0 have a similar number of individuals, and Cluster 2 is less than half the size. This result is also expected since Sweet and Dessert Wines are the least preferred among all the individuals. And Exotic wines also have a low average percentage in the initial data frame. So, if we have a cluster that normally prefers those types of wines more than the dry ones – or at least more than the average – this cluster should be significantly smaller than the others.

4.4 Introducing cluster to the original Data Frame and Final Conclusions

Finally, we introduced the clusters to our original data frame (after removing the outliers). This will allow us to examine the original data without standardization.

km_labels_3wh	Custid	Dayswus	Age	Educ	Income	Kidhome	Teenhome	Freq	Recency	Monetary	LTV	Perdeal	Dryred	Sweetred	Drywh	Sweetwh	Dessert	Exotic	WebPurchase	WebVisit	Access
0.000000	5996.44	900.83	50.46	17.54	73575.07	0.37	0.73	14.77	57.64	609.42	180.71	31.87	74.05	2.47	18.60	2.45	2.37	11.57	44.91	5.67	0.34
1.000000	6003.89	897.01	51.85	16.79	75725.16	0.41	0.34	17.30	61.66	777.33	287.10	27.96	41.58	6.84	38.40	6.52	6.65	13.32	37.14	4.57	0.22
2.000000	5999.90	893.41	35.29	15.05	51255.93	0.55	0.22	8.92	72.71	334.99	107.93	42.39	20.67	16.43	28.56	17.76	16.58	30.54	47.63	5.58	0.05

Fig. 14 – Segmentation 2 – average values

<u>Conclusions among 'Wine features'</u>

As a result of reintroducing the original data frame, our previous conclusions remain valid and consistent, and our clusters also are well segmented across the non-chosen features. Let us compare cluster by cluster, in detail:

- Firstly, please note that the average percentage for each wine feature by cluster represents the average percentage of individuals within the cluster among other wine types and not among other clusters. And it does not represent the average of the cluster spending (monetary). Different individuals within the cluster have different spending behaviours, but we are averaging individuals equally, independent of other features.
- **Cluster 0**: As previously noticed in different analysis, those are the ones with higher preference for Dry Red Wine (74.05% on average vs. 50.38% for the entire data frame), combining with White Dry Wine, these categories make up 93%. This cluster has a clear preference, and it seems to reject or rarely include sweet or dessert wines in their purchases (approx. 7%). They occasionally buy Exotic wines (11.57%), but less frequently than the overall average (16.54%).
- **Cluster 1**: Again, this cluster is the most balanced when comparing with the entire data frame. Sweet/Dessert wines represent each a bit more than 6.5% of their consumption, totalling an average of 20% of their wines purchases. On the other hand, they tend to prefer more White Dry Wine than the average (38.4% vs 28.52%), but less Red Dry wine than the (41.58% vs. 50.38%). Interestingly having the same 'balance' as the entire data frame, when comparing the consumption of dry vs. sweeter wines (80%/20%). It shows similar behaviour to Cluster 0 for Exotic wines.
- **Cluster 2**: With the non-normalized data, we see that this cluster has an equal balance between Dry and non-dry wines (49.23% vs. 50.77%). Even being the ones who purchase the most non-dry wines, it still represents only half of the wines purchased. They prefer Dry White over Dry Red but seems to have equal preference when comparing Sweet White, Sweet Red and Dessert Wines (around 17%). As also stated before, they are the most enthusiastic of exotic wines among all clusters (30.54%).

5. Final clustering

5.1 Hierarchical clustering

Merging the two segmentations produces a matrix of 12 clusters, all with statistically significant representation (ranging from 204 observations to 1259). Of these clusters, we wish to combine the most similar ones to reduce the number of clusters and have simpler and more readable results.

The hierarchical clustering method helps us define the right cut for the final clustering. Looking at the chart, we opt for 4 final clusters, having also performed some simulations with ± 2 and ± 1 clusters just to compare the results and checking any possible overlapping of the clusters.

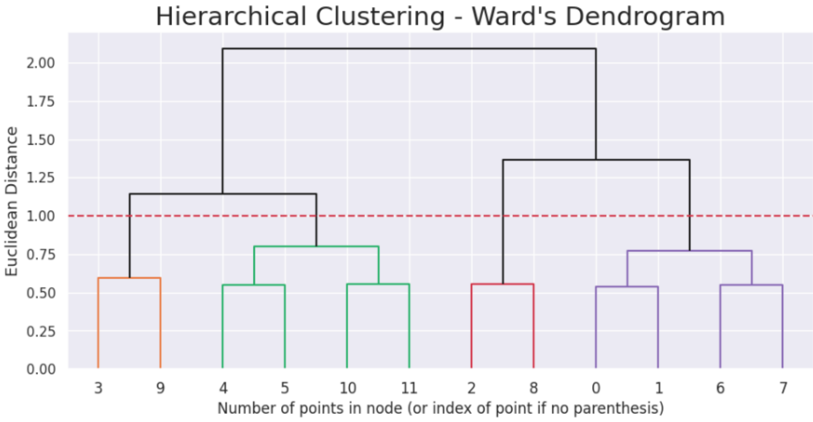

Fig. 15 – Hierarchical clustering

We can now reclassify the 12 clusters under 4 according to the dendrogram.

Before going more in depth in the cluster visualization according to our preferred choice, there are two further steps that need to be mentioned at this stage, although they are performed later after the number of final clusters is confirmed.

The first one is the recovery of the original scaling, which allows us to use the values of the cells as absolute values. This is particularly important for percentages, since the normalized values can create confusion and give the impression that the 0-1 normalized values are the real percentages in decimal scale.

The second step is the creation of a decision tree that tests the precision of our model in classifying the records under the right cluster. Our results show that around 90% of the customers are classified correctly, which is satisfactory. The decision tree will also allow us to re-classify the outliers that were trimmed during data preparation, in order to have each record of the dataset assigned to a cluster.

5.2 Cluster visualization

The line chart shows some overlapping between clusters for certain variables, but no cluster is too similar to the others. There are always variables that clearly differentiate between clusters, in either wine-related and engagement-related variables. We also notice that Recency and Dayswus are not relevant to any segmentation.

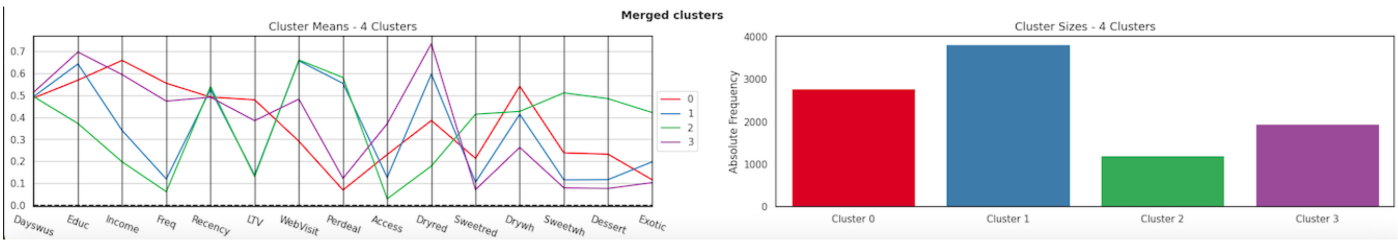

Fig. 16 – Final segmentation – line and bar chart

The heatmap of the final clustering reinforces the idea that each of our four final clusters shows different patterns. Those clusters that are more similar in the engagement segment differentiate well in the wine segment, and vice-versa.

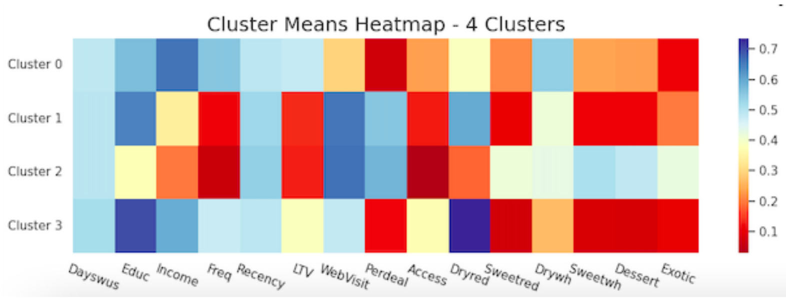

Fig. 17 – Final segmentation – heatmap

5.3 Cluster analysis

We should now have a closer look at each cluster, their features and what they represent as a whole. Naming the clusters with a catchphrase helps remember them easily.

hc_merged_labels	Dayswus	Age	Educ	Income	Kidhome	Teenhome	Freq	Recency	Monetary	LTV	Perdeal	Dryred	Sweetred	Drywh	Sweetwh	Dessert	Exotic	WebPurchase	WebVisit	Access
0.000000	892.00	64.90	16.60	96186.80	0.10	0.30	26.00	50.30	1233.50	485.40	6.60	38.80	8.50	36.10	8.40	8.10	9.80	22.80	2.90	0.30
1.000000	896.50	38.70	17.10	54452.20	0.70	0.60	6.40	70.20	183.40	10.30	53.30	59.50	4.20	28.10	4.00	4.10	16.80	54.50	6.60	0.10
2.000000	896.00	25.30	15.00	35770.00	0.70	0.20	3.80	81.40	74.80	1.20	55.80	18.50	16.70	28.90	18.40	17.50	35.90	56.40	6.60	0.00
3.000000	910.40	58.30	17.60	87694.10	0.10	0.60	22.40	50.20	1005.80	353.90	11.70	73.10	2.80	18.60	2.80	2.70	8.70	36.40	4.80	0.50

Fig. 18 – Final segmentation – average values

<u>Cluster 0 - "White hair, white grapes" - Affluent seniors, white wine oriented</u>

Status: The group has the highest income (€ 96K), age and LTV. Low sensitivity (6,6) to discounts supports the wealthy status, the age explains the low digital competences, given the lowest web visits. Age is also coherent with the very low number of observations with minors at home - these are probably retired individuals and/or grandparents. They tend to buy often (Freq = 26) and are not particularly interested in accessories (0,30).

Wines: While having a good balance between dry red (38,8%) and dry white wines (36,1%), they show a clear higher consumption of white wine relatively to the other segments.

Size: The cluster is the second biggest after Cluster 1 (2775 observations).

<u>Cluster 1 - "Family Dry" - Family-oriented emerging, dry wine lovers</u>

Status: A group with mid-low income (€ 54 K) but high education (17,1 years), which would suggest younger people with bright perspectives. The cluster is family-oriented, with a clear majority of teens and/or kids at home. Variables show affinity to digital competences with high web visits (6,6) and sensitivity to discounts (53,3). They do not buy frequently (6,4) and are not interested in accessories (0,13).

Wines: The group has a more balanced basket between dry red (59,5%) and dry white (28,1%), compared to the other groups, with a preference for dry wines (close to 90% of the total).

Size: They are the most represented cluster in our sample (3830 observations).

Cluster 2 - "Poor sweethearts!" - Low income and education, sweet wine affinity

Status: The cluster has the lowest education level (15,0 years) and lowest income (ca. € 36 K), with a tendency to buy on the web (6,6) and on discounted products (55,8). The group is very young, as shown by the high presence of kids at home, but not teenagers. They do not buy frequently (3,7) and are the least interested in accessories (0,03).

Wines: The group shows a higher interest for sweet and dessert wines (51% of the total), but the combination of this with low LTV might suggest that sweet wines are more represented in the average basket because they buy little wine in general, so the few purchases of sweet wines have a higher impact on the basket.

Size: They are the least represented cluster in our sample (1207 observations).

Cluster 3 - "Grades of red" - Wealthy educated families, dry red wine lovers

Status: The group has the highest education (17,6 years), and still a high age and LTV. They still use the web to make purchases (4,8) but are not particularly sensitive to discounts (11,7). They have teens at home rather than kids, which suggests the parents are at least in their 30s, probably already in their 40s, starting to have increasing purchasing power due to their education and income, and children about to leave the household. They tend to buy often (Freq = 22,3) and are the segment that shows the most interest in accessories (0,4).

Wines: They have a clear preference for dry red wines (73,11%).

Size: The cluster is the second lowest after Cluster 2 (1943 observations).

6. Business insights

In this section we propose a concise list of insights and business advice.

The first part describes how the marketing should be done on the different clusters, and how their evolution should be followed.

The second section tackles how the different products and channels should be developed and where the efforts should be placed.

Recommendations for cluster marketing

- Cluster 0 - White hair, white grapes (Affluent seniors, white wine oriented)

This cluster is profitable with a high LTV, however given the average age this is the cluster that includes customers with the shortest remaining life expectancy. It would be interesting to analyse into detail whether the wine basket preferences are related to age or rather to specific generations. In the first case, dry wines might still be the most interesting ones in the future for older customers. On the other hand, should the wine preferences depend more on generational trends, this cluster might gradually shift more towards dry red wine.

- Cluster 1 - Family Dry (Family-oriented emerging, dry wine lovers)

The most represented cluster must be followed with particular attention. The individuals in this cluster, which includes well-educated people with good future income perspective, have the potential to shift closer to Cluster 3, therefore with very profitable future perspectives. The relatively balanced basket between red and white dry wines suggests that a balanced approach should be kept in the future product strategies for wines. Should a specific group be targeted with promotions and aggressive campaigns, it should be this one, since it has a high volume of individuals, it is sensitive to pricing and has a longer purchasing perspective.

- Cluster 2 - Poor sweethearts! (Low income and education, sweet wine affinity)

Our least represented and least profitable cluster still can provide some interesting insights and challenges. Without doubt this is the least important cluster, however one important question should be raised. Is the low income and LTV level in this cluster only temporary or not? If further analysis would indicate that this is the case (e.g this cluster is still in education phase) we would have a potential for improvement in the future and could see many of the individuals currently represented in this cluster shift towards the Cluster 1 and potentially later even to Cluster 3. On the other hand, if the income and LTV levels are not expected to increase significantly, there will be no improvements in profitability to be expected and this cluster will remain a low-value segment for the business. At the moment, the interest shown by this segment to sweet wines is not enough to justify any important investment in these product segments. Any change in strategy should be targeted at keeping the price low for sweet wines.

- Cluster 3 - Grades of red (Wealthy educated families, dry red wine lovers)

This is a relatively low volume but highly profitable segment, which could guarantee business continuity by the presence of teenagers in the household as potential future customers. The high levels of income, LTV and education suggest that this cluster would value new high-quality dry red wines, which could be a product segment to invest in.

To sum up, our most important clusters are Cluster 0 and Cluster 3, both with high LTV and with relevant sizes, especially for Cluster 0. Cluster 1 is a customer segment to invest on, given the size, the age and the future potential shown by demographics. Cluster 2 is by far the least relevant, both in size and LTV.

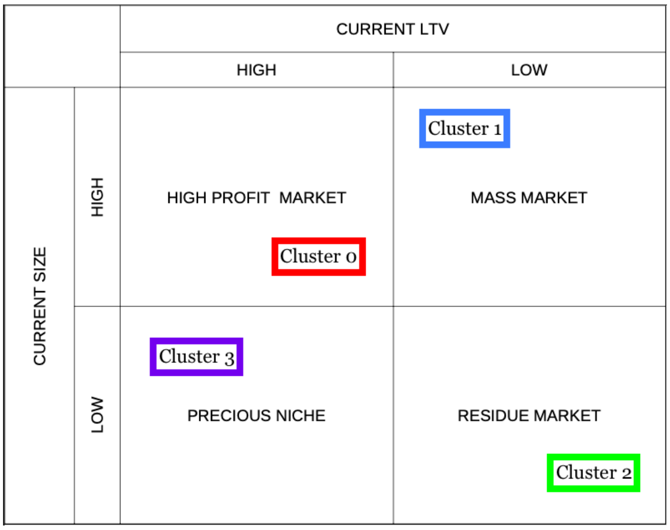

Fig. 19 – Cluster importance matrix

<u>Recommendations for product management</u>

Wines:

Emphasis should be given to dry wines, supported by Cluster 0 and Cluster 3. One important trend to follow, will be the evolution of preferences between dry red and dry white wines. Whether these are generational trends or more related to demographical factors such as income and age, this will affet the relative proportions in the future and influence our business strategy.

Dry white wines might drop in the future due to the high age of Cluster 0, but we cannot exclude that age will be a factor that contributes to shift the preferences from other clusters - Cluster 3 above all.

Sweet wines - red and white - as well as dessert wines are only preferred by our least favourite cluster, Cluster 2. This suggests that little investment should be done on them.

Exotic wines are not particularly favoured by the customers, however the interest shown by Cluster 1 is not negligible. We should follow the evolution of Cluster 1 and see how this relates to the purchase of exotic wines.

Accessories:

Cluster 0 and Cluster 3 should be targeted with accessories. Cluster 3 seems to already have a certain degree of awareness on the accessories, whereas Cluster 0 might need some different targeting and marketing campaigns due to the different purchasing channels used and the low degree of web purchases.

Discounts:

Discounts should only be targeted at Cluster 1 and 2, those that are sensitive. The discounts are more important to clusters that do web purchases with consistency.

Website:

The website accesses will be important in the future, since they are key features for the youngest clusters, and we can assume that those habits will not change with time. We additionally suggest targeting discount mainly on the online store, rather than in the physical stores.

Annex

[1]

Cramér's V is an effect size measurement for the chi-square test of independence. It measures how strongly two categorical fields are associated.

The effect size is calculated in the following manner:

- Determine which field has the fewest number of categories.
- Subtract 1 from the number of categories in this field.
- Multiply the result by the total number of records.
- Divide the chi-square value by the previous result. The chi-square value is obtained from the chi-square test of independence
- Take the square root.

Interpretation of Cramers V:

- ES ≤ 0.2 : The result is weak. Although the result is statistically significant, the fields are only weakly associated.
- 0.2 < ES ≤ 0.6 : The result is moderate. The fields are moderately associated.
- ES > 0.6 : The result is strong. The fields are strongly associated.

The interpretation of the effect size (ES) is subject to the field of study.

[2] **Point-Biserial Correlation**

The Point-Biserial Correlation is a special case of the Pearson Correlation (If you treat the binary variable as continuous the computation is mathematically equivalent to Pearson correlation). Is used when you want to measure the relationship between a continuous variable and a variable that has two values (binary variable).

Interpretation

- The value also ranges from −1 to +1, but it specifically represents how the mean of the continuous variable differs between the two groups defined by the binary variable.
- A higher absolute value indicates a stronger relationship between the binary grouping and the continuous variable.